

CSE 321 Database Systems

Lab Session #2 (2026 Version)

Relational Schema Design and Normalisation Report

Overview

In this lab session, you will convert a given ER diagram into a relational database schema. The purpose of this activity is not only to practise relational schema design, but also to develop your ability to analyse functional dependencies, identify normalisation issues, and justify schema design decisions in a precise and logical manner.

This lab is designed as an individual exercise. You are expected to complete the schema design task during the lab session and later prepare a short reflective report based on your own submitted work.

Deadline

The submission deadline is **29-May, 23:59**. Please submit your assignment in advance to avoid any last-minute issues. Late submissions will receive a score of 0.

In-Class Activity

During Lab Session #2, an ER diagram will be provided in class. Based on that diagram, you must construct a relational schema.

You should aim to produce a schema that is as complete, consistent, and well-structured as possible. Each relation should include clearly defined attributes and keys.

Attending the in-class lab session activity is mandatory.

Example

$R1(A, B, C, D)$

Keys: $\{A, B\}, \{A, D\}$

PK: $\{A, B\}$

FDs:

$AB \rightarrow CD, \quad D \rightarrow B$

Relation R1 is in 3NF because:

- $AB \rightarrow CD$ is OK, since AB is a key of R1.
- $D \rightarrow B$ is OK, since B is a prime attribute, i.e., B is included in a candidate key $\{A, B\}$.

After the Lab Session

After the lab session, I will collect all submissions and scan them. The scanned version of your own submission will then be distributed back to you individually.

At that point, you must carefully review your original schema again. If necessary, you may also use LLM tools for assistance. However, the purpose of this step is not simply to generate a correct answer, but to analyse your own design process critically and transparently.

You should identify:

- what you designed correctly,
- what you designed incorrectly,
- whether your schema satisfies 3NF,
- what functional dependencies you missed or misinterpreted,
- what assumptions you made,
- whether those assumptions were reasonable,
- how and why you revised the schema.

Report Submission

You must prepare an individual report in English using Overleaf. I will upload the report template separately, and you must use that template for your submission.

The report must be written in a clear and organised manner, with a maximum length of 25 pages.

Reasonable assumptions are allowed when the ER diagram does not fully specify all semantic constraints. However, all assumptions must be stated explicitly and justified clearly.

Your report must include all of the following components:

- the original schema that you submitted during the lab session,
- the revised schema,
- a clear specification of keys, PK, and functional dependencies for each relation,
- a justification of whether each relation satisfies 3NF,
- a detailed explanation of each modification,
- a discussion of any misunderstanding, omission, or incorrect assumption.

A good report is expected to do more than present two schemas. It should clearly explain the difference between the original and revised versions, and demonstrate that you understand why the revision was necessary.

Expected Report Structure

You may organise your report in any reasonable way, but the following structure is strongly recommended:

1. Introduction: Briefly explain the purpose of the report.
2. Review of the Original Design: Present the exact schema you submitted during the lab session. Then, explain your reasoning, including how you derived relations, keys, functional dependencies, and assumptions.
3. Revised Design: Present your fully corrected schema. For each relation, clearly specify the attributes, keys, primary key, functional dependencies, and assumptions. Also explain whether each relation satisfies 3NF.
4. Revision Details: For each modification from the original design, explain (1) what was wrong in the original design; (2) why it violates relational design principles (e.g., 3NF); and (3) how the revised schema resolves the issue.
5. Conclusion: Summarise what you learned from the exercise.

Evaluation

The grading criterion for this lab is intentionally simple.

- If the report is written clearly and demonstrates careful reflection and explanation, it will receive 4/4.
- If the report is incomplete, unclear, or lacks sufficient explanation, it may receive 3/4 or lower.
- If you do not attend the lab activity session, or if you submit after the deadline, the score will be 0.

In other words, this lab is primarily evaluated based on the clarity, completeness, and sincerity of your reflective report, not solely on whether your original schema was perfect.

Final Remark

This lab is intended to help you develop an important skill in database design: not only constructing a relational schema, but also reviewing your own design critically, identifying where your reasoning failed, and explaining how to improve it in a structured way.

The purpose of this report is not to hide mistakes, but to demonstrate that you can analyse and improve your own design critically and logically.

Using LLM tools to generate the entire report without genuine reflection will be considered inappropriate and may result in a reduced score.

A correct schema is valuable, but a clear and honest explanation of your design process is equally important.